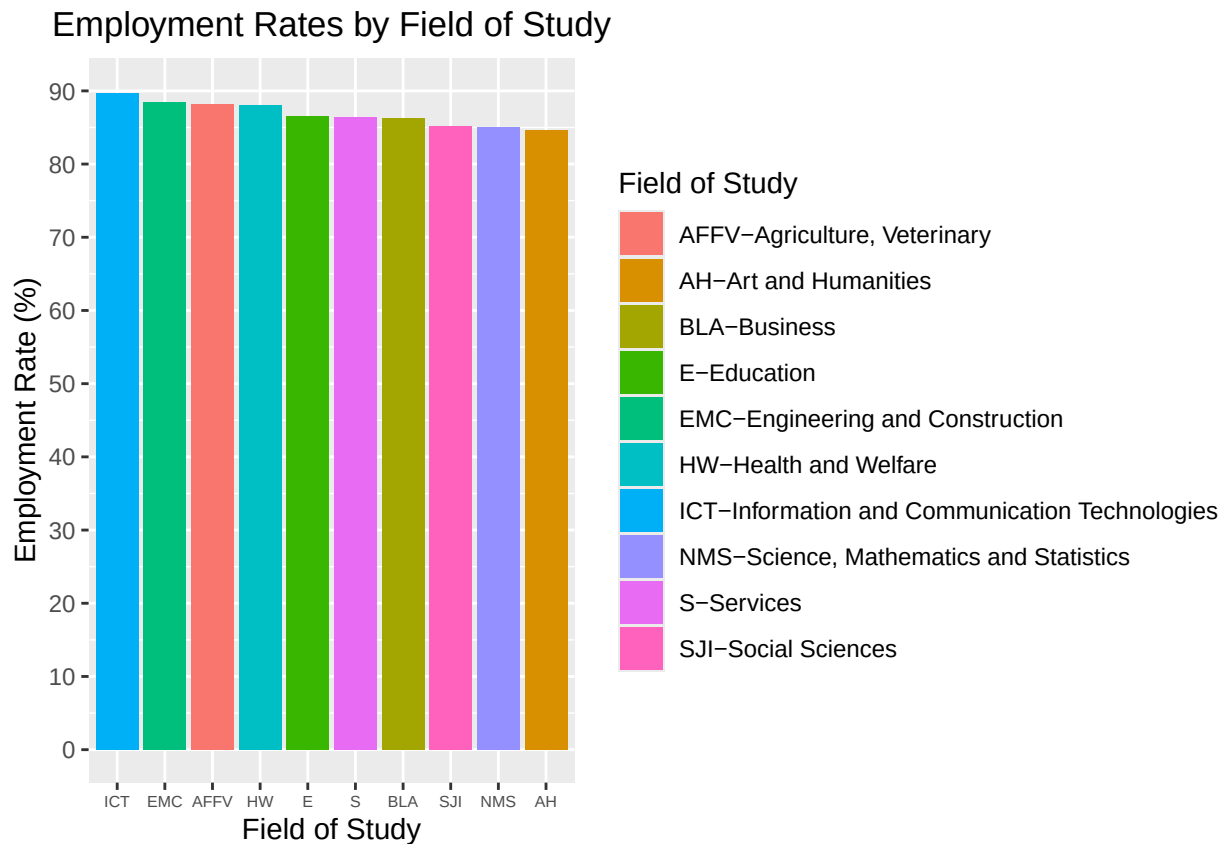


Assignment 8:The Impact of University Majors on Graduate Success: Employment and Earnings

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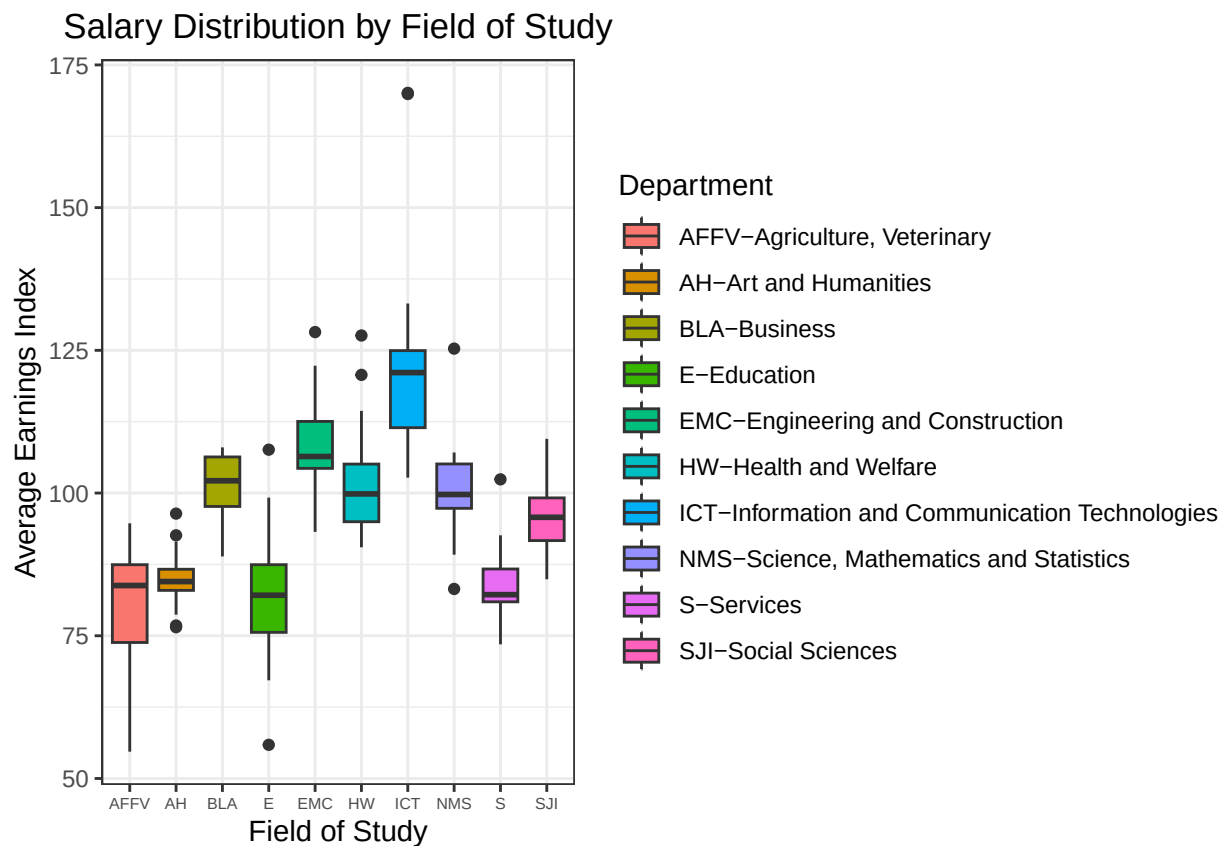
```
ggplot(employment_avg, aes(x = reorder (Department, - mean_rate) , y = mean_rate , fill = Department)) +  
  geom_bar(stat = "identity", position = "dodge") +  
  labs(x = "Field of Study",  
       y = "Employment Rate (%)",  
       fill = "Field of Study",  
       title = "Employment Rates by Field of Study") +  
  scale_fill_discrete(labels = c("AFFV-Agriculture, Veterinary", "AH-Art and Humanities", "BLA-Business",  
    "EMC-Engineering and Construction", "HW-Health and Welfare", "ICT-Information and Communication Technologies",  
    "NMS-Science, Mathematics and Statistics", "S-Services", "SJI-Social Sciences")) +  
  scale_y_continuous(limits = c(0, 90) , breaks = seq(0,90,10)) +  
  theme(plot.title = element_text(hjust = 0.5)) +  
  theme(  
    axis.text.x = element_text(size = 6))
```



When we look at this graph, we can see Information and Communication Technologies has the highest employment rate. The proportion is almost 90%. Art and Humanities has the lowest employment rate at the same time. We would think people usually choose Information and Communication Technologies.

```
ggplot(earning_long, aes(x = Department, y = Earnings, fill = Department)) +
  geom_boxplot() +
  labs(
    title = "Salary Distribution by Field of Study",
    x = "Field of Study",
    y = "Average Earnings Index"
  ) +
  scale_fill_discrete(labels = c("AFFV-Agriculture, Veterinary", "AH-Art and Humanities", "BLA-Business",
    "E-Education", "EMC-Engineering and Construction", "HW-Health and Welfare", "ICT-Information and Communication Technologies",
    "NMS-Science, Mathematics and Statistics", "S-Services", "SJI-Social Sciences")) +
  theme_bw() +
  theme(plot.title = element_text(hjust = 0.5)) +
  theme(
    axis.text.x = element_text(size = 6))
```

```
## Warning: Removed 12 rows containing non-finite outside the scale range
## (`stat_boxplot()`).
```



When we look at this graph, salaries are not balanced in the departments. Some departments earn much but some departments not that much. While salaries are tighter in some departments, they have a wider range in others. We first look one by one.

Information and Communication Technologies and Engineering and Construction. 'Information and Communication Technologies' and 'Engineering and Construction' has the highest earnings. But they have a problem. That is variability problem. variance too high. That means, there are large differences in people's earnings.

Business. Average earnings is high other departments (not ICT or Engineering and Construction). And that's variance not too highest. For that reason we can say more people earn similar earnings.

Scines, Mathematics and Statistics. Variance not too high or too low. For that reason, there are not large differences in peoples's earnings like ICT or Engineering and Construction or not low differences in people's earnings like Business. So, we can say that, this is more balance.

Health and Welfare looks like Business and Science, Mathematics and Statistics. But this is more balance than the other two.

Social Scines's median is lower to intermediate. They earns less.

Art and Hummanities's median is lower. They earns less but earns may be closer each other.

Services's median is lower. They earns less and earns closer each other. Distribution is too homogeneous.

Education's medians is lower. They earns less and earns closer each other.

Agriculture and Veterinary's median is lower but variances not too lower. For that reason some people's earns more high but not all of them.

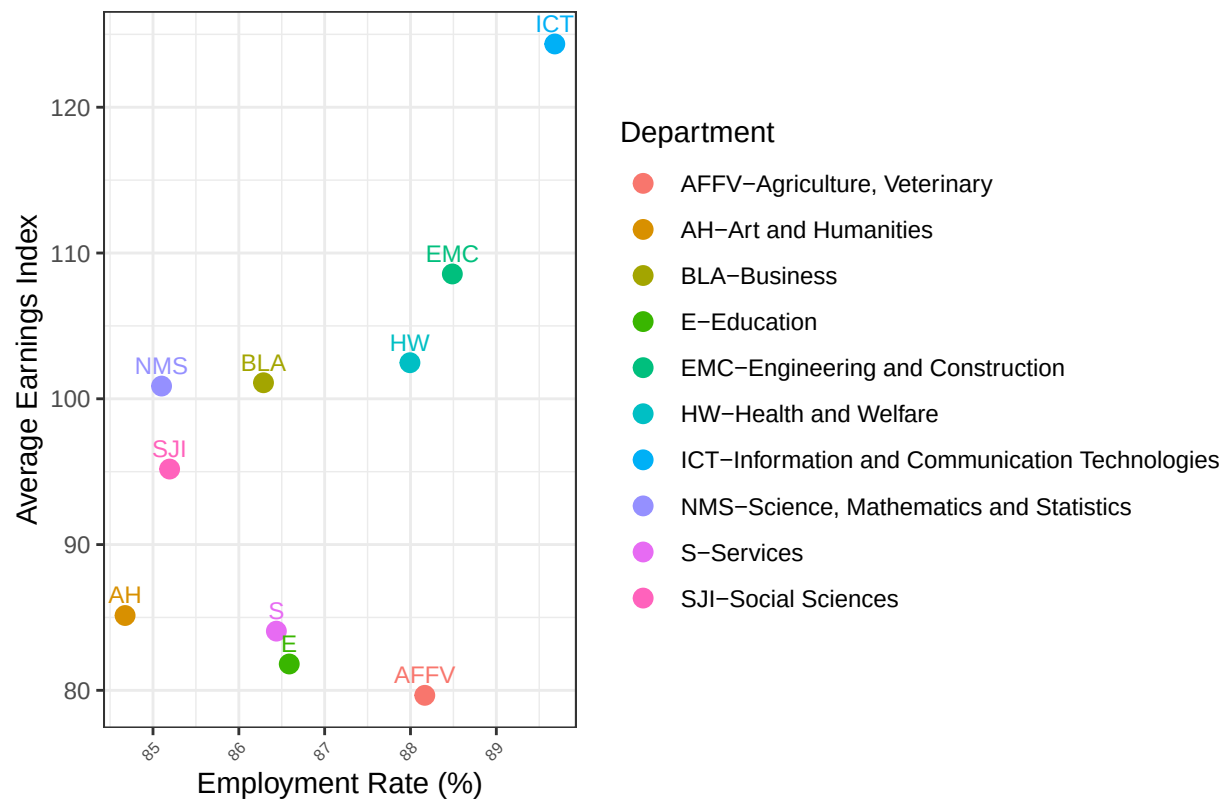
In shortly; High Earning Domains ICT-EMC-HW ICT and EMC's median and variances values range are wide. But HW more a balanced. This difference for in ICT and EMC that could be mean technical skills, experience.

Mid Earning Domains BLA-NMS-SJI Variances differences are important. BLA: There are inequality is most pronounced. So, Variance too high. NMS: most balanced. Variance not too high or low. SJI: Median is lower but variance is balance.

Low Earning Domains AH-S-E-AFFV AH: Lower earnings and lower variances. S-E: Lowest earnings and lower variances. Most homogeneous. AFFV: Median is lower.

```
ggplot(earn_employment, aes( x = Employment_avg , y = Earnings_avg , color = Department)) +
  geom_point(size = 3) +
  geom_text( aes(label= Department),
             vjust = -0.7, size = 3,
             show.legend = FALSE) +
  labs(
    title = "Relationship Between Employment and Earnings",
    x = "Employment Rate (%)",
    y = "Average Earnings Index",
    color = "Department"
  ) +
  scale_color_discrete(labels = c("AFFV-Agriculture, Veterinary", "AH-Art and Humanities", "BLA-Business",
    "E-Education", "EMC-Engineering and Construction", "HW-Health and Welfare", "ICT-Information and Communication",
    "NMS-Science, Mathematics and Statistics", "S-Services", "SJI-Social Sciences"))+
  theme_bw() +
  theme(plot.title = element_text(hjust = 0.5) ,
        axis.text.x =element_text(angle = 45, hjust = 1)) +
  theme(
    axis.text.x = element_text(size = 6))
```

Relationship Between Employment and Earnings



This plot shows a positive relationship between employment rates and earnings. In short, if departments have high employment rates, they generally have high earnings.